

LOAN APPROVAL PREDICTION SYSTEM

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Introduction to the Domain (AI / ML in Finance)

- Rapid digital transformation in banking
- Need for automation in loan approval
- Manual process → slow, subjective & error-prone
- ML enables fast, fair & data-driven loan decisions



PROBLEM STATEMENT

High volume of loan applications

Manual evaluation takes time & varies between officers

Need for:

- ✓ automation
- ✓ accuracy
- ✓ consistency
- ✓ fairness

Goal: Build ML model to predict loan approval status



OBJECTIVES OF THE STUDY

1. Develop ML-based loan approval system

Create a machine learning model that can automatically decide whether a loan should be approved or rejected based on applicant data.

2. Preprocess & analyze applicant data

Clean the dataset, handle missing values, convert categories, scale numbers, and study patterns (EDA) to improve model accuracy.

3. Compare multiple ML models

Train and test different algorithms like Logistic Regression, Decision Tree, and Logistic Regression to find which performs best.

4. Deploy best-performing model using Gradio

Convert the best model into a simple web interface using Gradio so users can input details and get predictions instantly.

5. Provide real-time predictions for banking staff

Allow bank officers to quickly check loan approval status in real-time, reducing manual effort and speeding up the decision-making process.

LITERATURE REVIEW – TRADITIONAL METHODS

- **Logistic Regression widely used in credit scoring**
- **Simple & interpretable**
- **Limitation: assumes linear relationships**
- **Used in several studies on UCI & Kaggle datasets**

LITERATURE REVIEW – ML ADVANCEMENTS

- **Decision Trees capture non-linear patterns**
- **Prone to overfitting**
- **Random Forest improves:**
 - ✓ **stability**
 - ✓ **accuracy**
 - ✓ **robustness**
- **Gap in previous studies: no complete automatic pipeline + deployment**

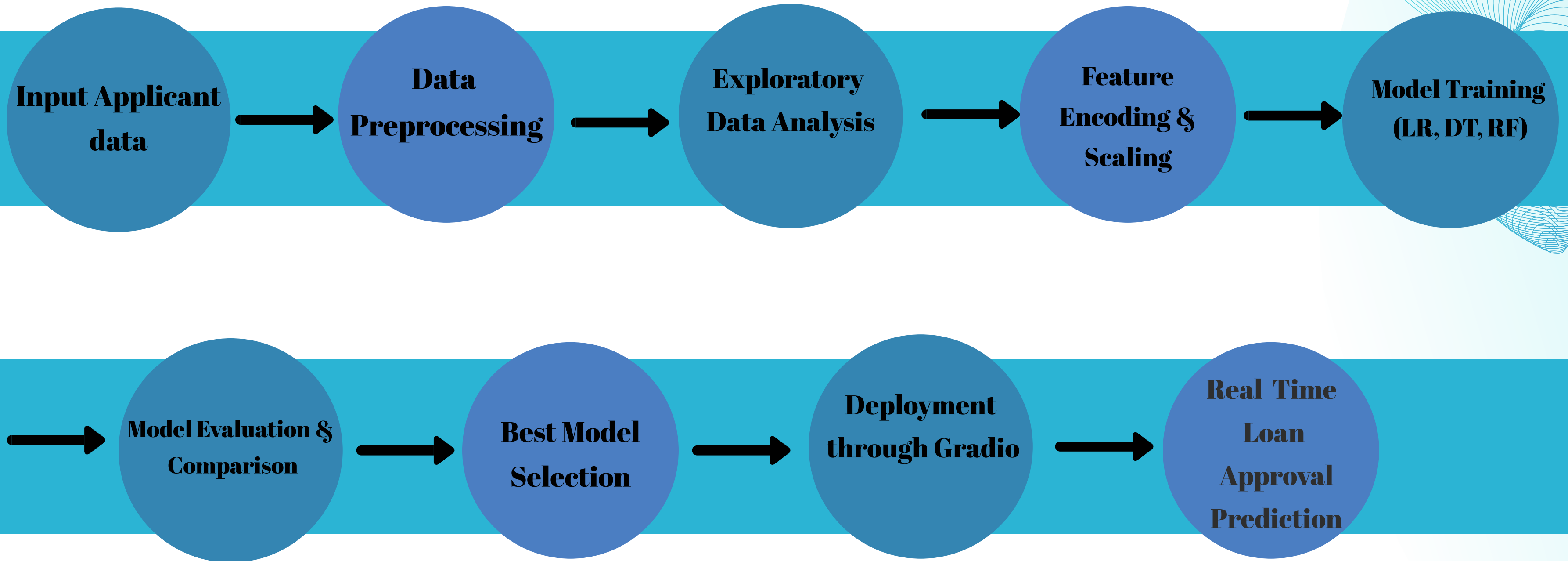
part 1

- **Logistic Regression in credit scoring**
- **Simple & interpretable**
- **Limitations: linear assumptions**

part 2

- **Decision Trees → non-linear modeling**
- **Random Forest → robust, avoids overfitting**
- **Prior work lacked complete pipeline & UI**

WORKFLOW OF PROPOSED SYSTEM



Methodology Overview

- Data collection & cleaning
- Exploratory Data Analysis (EDA)
- Preprocessing pipeline creation
- Training ML models
- Model comparison
- Deployment using Gradio



Exploratory Data Analysis (EDA)

- **Loan approval imbalance observed**
- **Credit history → strongest predictor**
- **Income vs Loan amount correlation**
- **Boxplots & histograms showed skewed income distribution**

Feature Importance

1. **Credit History**:- applicants with a good past repayment record are more likely to get loan approval.
2. **Loan Amount**:- Higher loan amounts increase the risk, so this significantly affects approval chances.
3. **Applicant Income**:- Higher income increases repayment ability, making approval more likely.

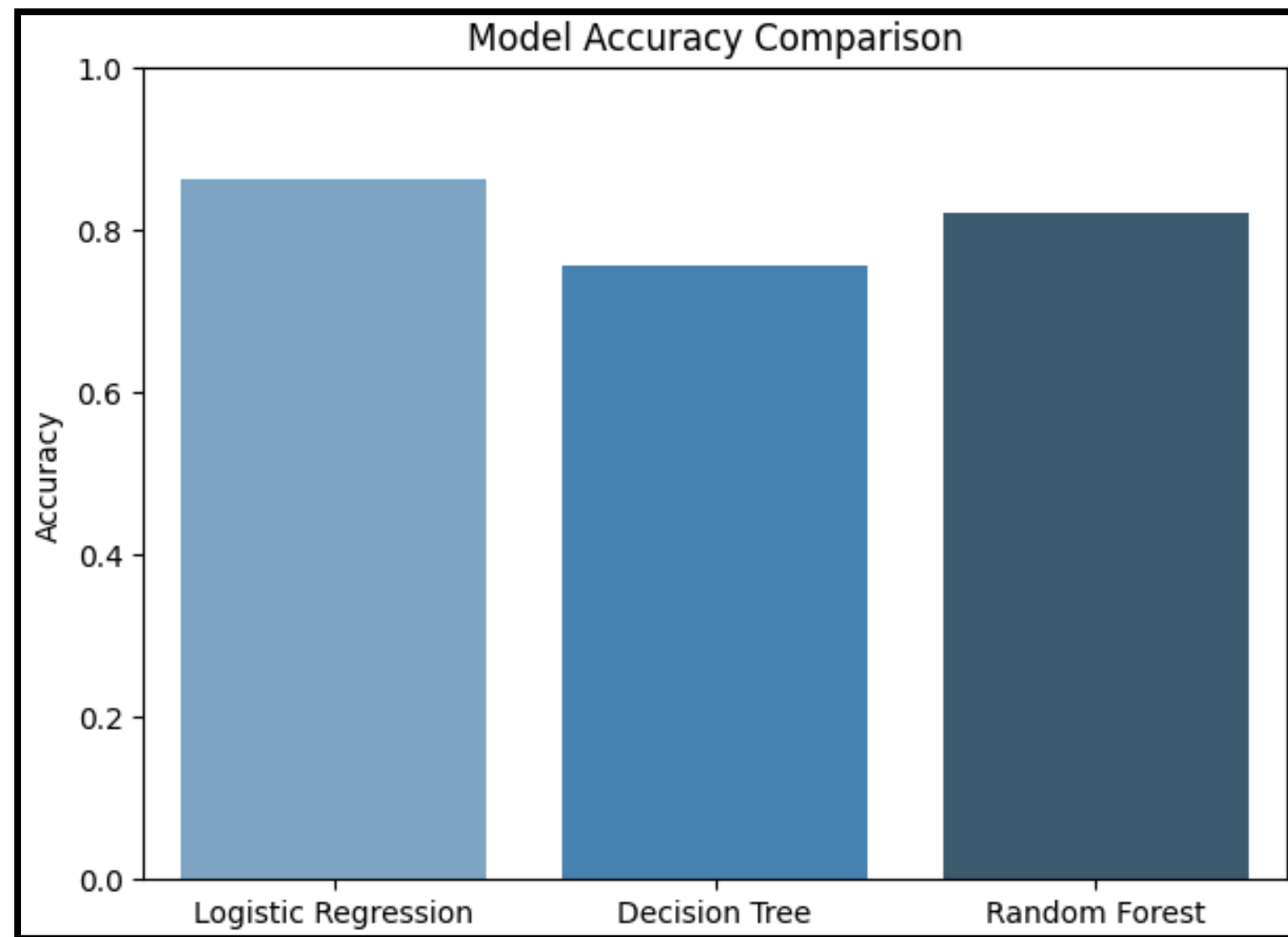
PREPROCESSING PIPELINE

- Numerical Features:
- Median imputation
- Standard scaling
- Categorical Features:
- Mode imputation
- One-hot encoding
- Combined using ColumnTransformer
- Ensures clean & structured input for all models

MODEL TRAINING & EVALUATION

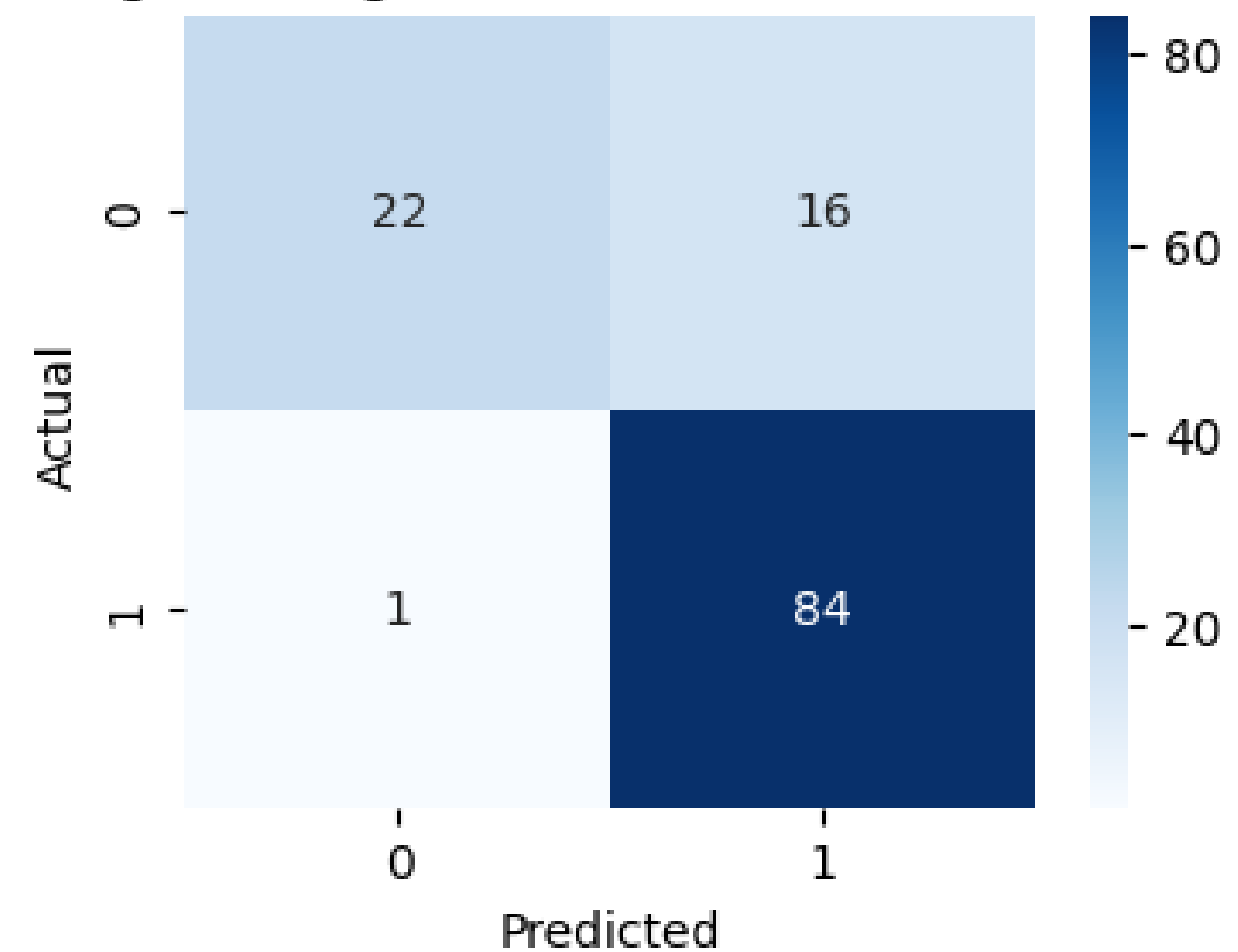
- Train–test split: 80–20
- Models used:
- ✓ Logistic Regression
- ✓ Decision Tree
- ✓ Random Forest
- Metrics: Accuracy, Precision, Recall, F1-score
- Logistic Regression performed best (~86%)

RESULTS

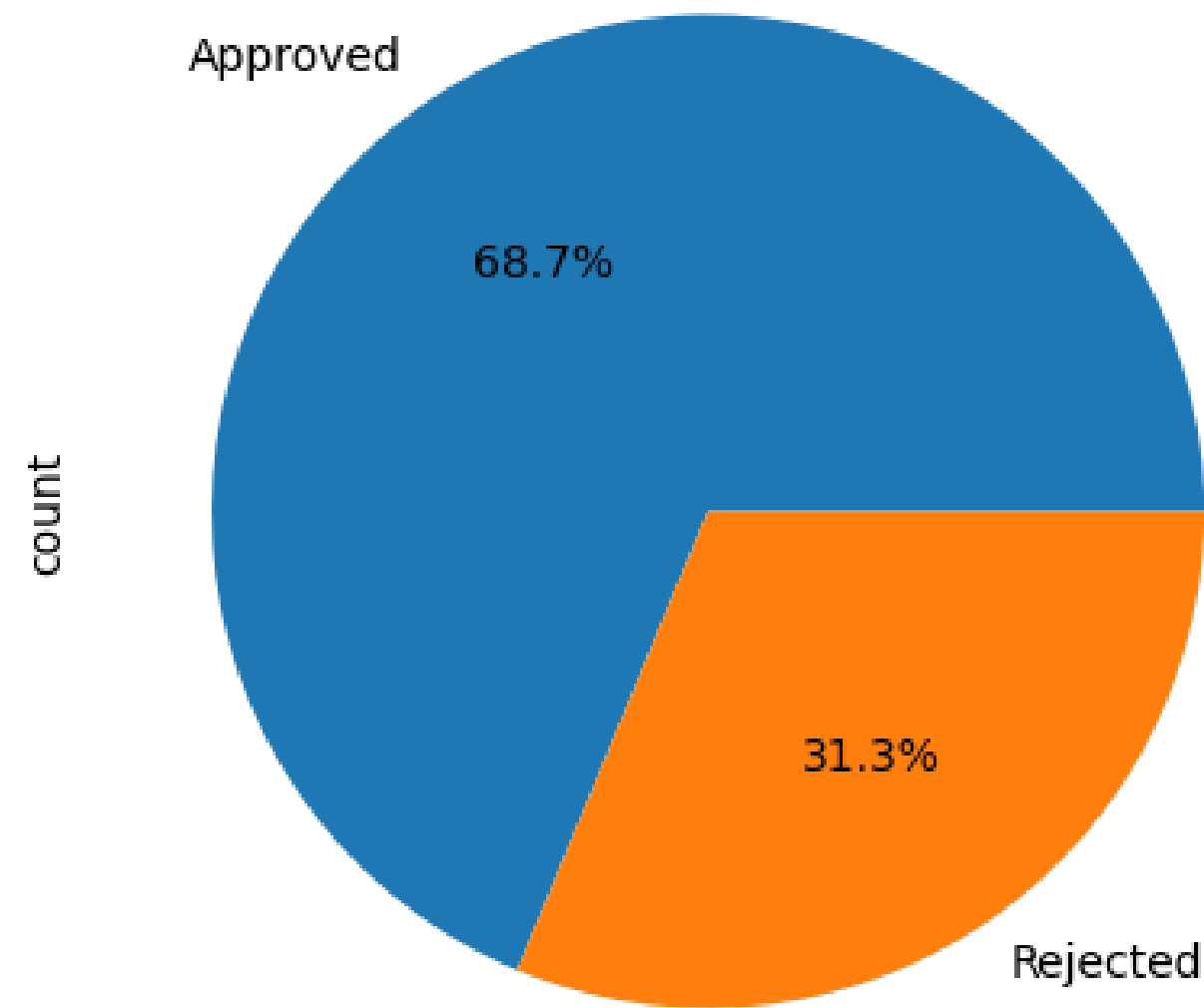


Logistic Regression performed the best with around 86% accuracy. Decision Tree had the lowest accuracy (~75%), and Random Forest performed moderately (~82%).

Logistic Regression — Confusion Matrix



The model correctly predicted 84 approved and 22 rejected loans. Only a few misclassifications occurred (16 false approvals and 1 false rejection). Overall accuracy $\approx 86\%$, with good precision and recall values.



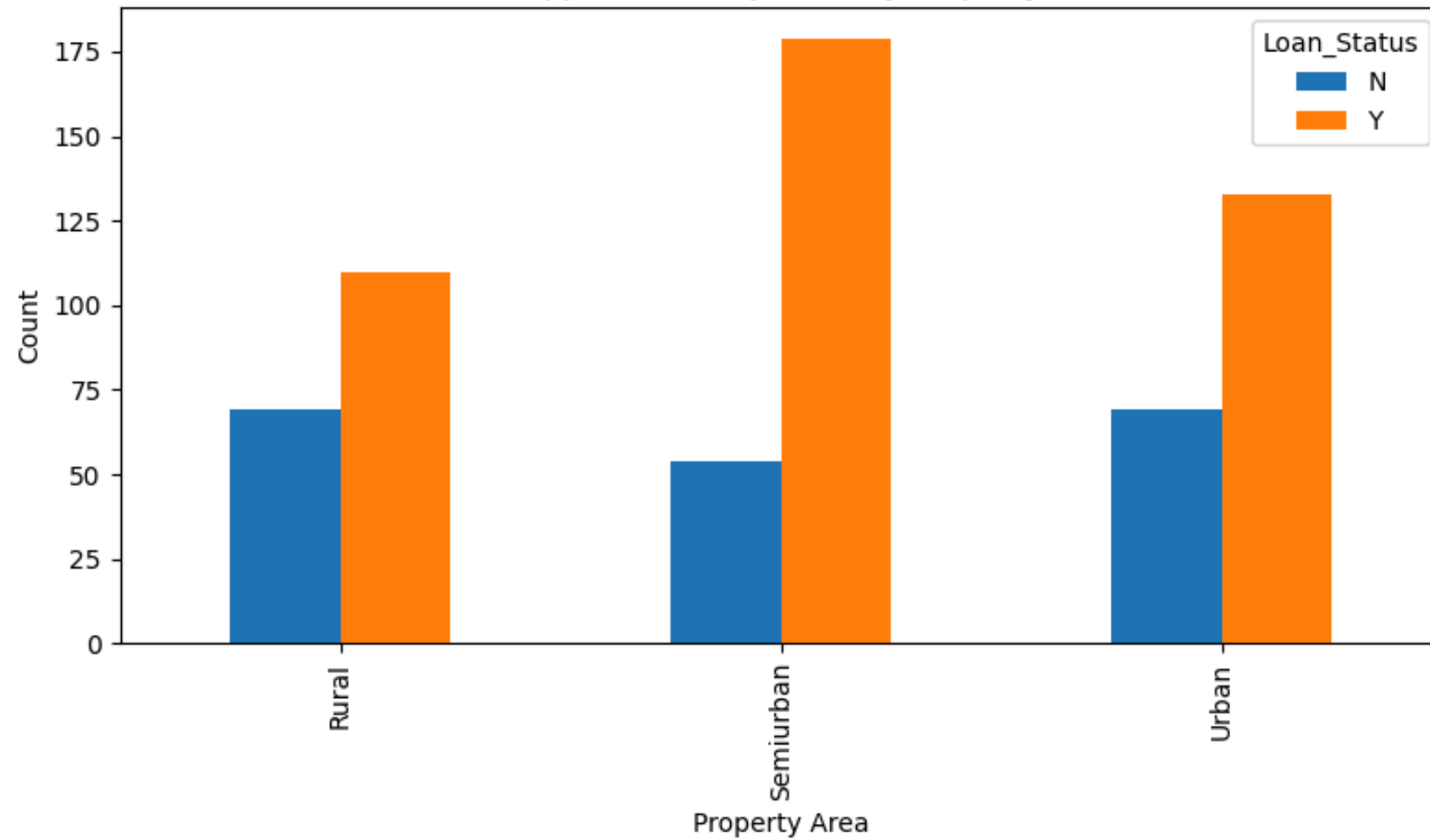
A pie chart showing the percentage of Approved and Rejected loans in the dataset

About 68.7% loans were approved. Around 31.3% were rejected.

ApplicantIncome	1	-0.12	0.57	-0.045	-0.015	-0.0047
CoapplicantIncome	-0.12	1	0.19	-0.06	-0.0021	-0.059
LoanAmount	0.57	0.19	1	0.039	-0.0084	-0.037
Loan_Amount_Term	-0.045	-0.06	0.039	1	0.0015	-0.021
Credit_History	-0.015	-0.0021	-0.0084	0.0015	1	0.56
Loan_Status_num	-0.0047	-0.059	-0.037	-0.021	0.56	1
	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Loan_Status_num

- Credit_History has a strong positive correlation with Loan_Status, meaning people with good credit history are more likely to get approved.
- ApplicantIncome and LoanAmount show a moderate relationship.
- Other features have weaker correlations.

Loan Approval vs Rejection by Property Area



- Semiurban = most approvals
- Urban = moderately high approvals
- Rural = good approval rate but lowest among the three areas
- Rejections are significantly lower in all areas compared to approvals

Loan Approved

Confidence: 66.68%

Risk Level: Medium Risk

Reasons for this decision:

Good credit history – indicates responsible repayment behaviour.
Sufficient combined monthly income for EMI repayment.
Loan amount is within a manageable range.
Property location (Urban) is favourable and has strong collateral value.
Graduate applicant – usually stable career prospects.
Salaried employment – income is more predictable.

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Loan Rejected

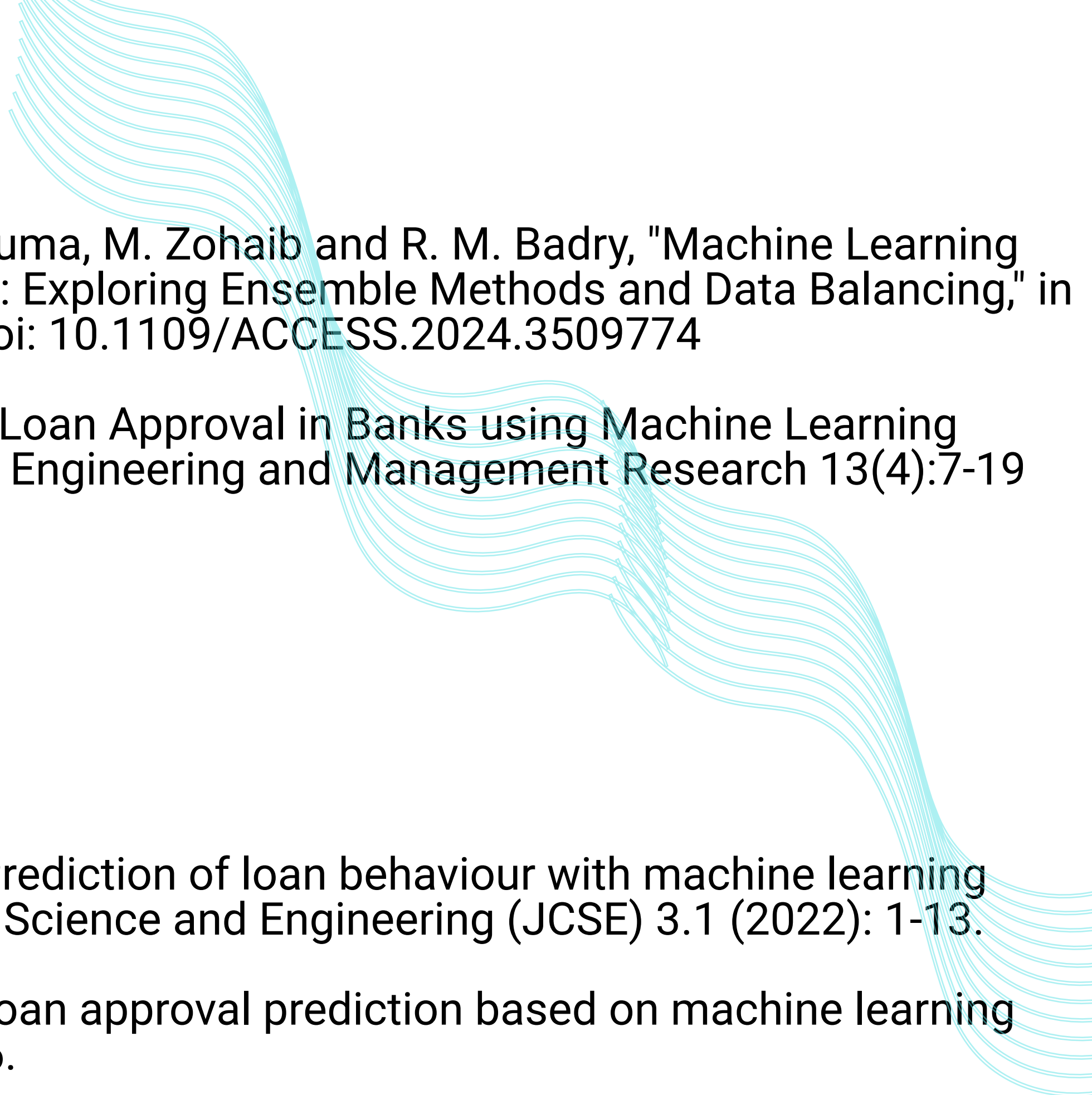
Confidence: 0.00%

Risk Level: High Risk

Reasons for this decision:

Good credit history – indicates responsible repayment behaviour.
Sufficient combined monthly income for EMI repayment.
Loan amount is within a manageable range.
Property location (Urban) is favourable and has strong collateral value.
Graduate applicant – usually stable career prospects.
Salaried employment – income is more predictable.

REFERENCES -

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CONCLUSION

- Successful development of automated loan approval system
- Logistic Regression was selected as the best model since it achieved the highest accuracy and stability compared to the others.
- Gradio makes the system usable in real-world banking



FUTURE SCOPE

- Larger datasets
- Deep learning models
- SHAP Explainability
- Cloud deployment for scalability



Thank you

